

Homework 2 Part 1 - Solutions

Exercise 1 (10 points)

Consider a training set containing positive natural numbers - including zero - ($x \in \mathbb{N}_0$) for 2 classes, C_0 and C_1 . The training set has 100 samples for class C_0 and 50 for C_1 .

Suppose that you have reason to believe that samples belonging from C_0 are drawn from a Poisson random variable with parameter $\lambda > 0$, and samples belonging to C_1 are drawn from a Binomial random variable with parameters $n \in \mathbb{N}_0$ and $p \in [0, 1]$. In other words, the data likelihoods for C_0 and C_1 are as follows:

$$p(x|C_0) = \frac{\lambda^x e^{-\lambda}}{x!}$$
$$p(x|C_1) = \binom{n}{x} p^x (1-p)^{n-x}$$

where $\binom{n}{x} = \frac{n!}{x!(n-x)!}$ is the binomial coefficient of " n choose x ".

Consider $\lambda = 2$, $n = 10$ and $p = 0.5$. Consider the test point $x = 3$. Use the Naive Bayes classifier to predict a class label for $x = 3$. Show your work.

The prior probability for each class can be estimated as its relative frequency in the training set, namely

$$p(C_1) = \frac{100}{150} = \frac{2}{3} \text{ and } p(C_2) = \frac{50}{150} = \frac{1}{3}$$

For the provided parameter values, the data likelihood for class 1 and class 2 are defined as:

$$p(x|C_0) = \frac{2^3 e^{-2}}{3!} = \frac{4e^{-2}}{3}$$
$$p(x|C_1) = \binom{10}{3} 0.5^3 (1-0.5)^{10-3} = \frac{120}{2^{10}}$$

For the test point $x = 3$, we compute the posterior probability with

$$P(C_k|x=3) = \frac{p(x=3|C_k)p(C_k)}{p(x=3)} = \frac{p(x=3|C_k)p(C_k)}{\sum_{j=1}^2 p(x=3|C_j)p(C_j)}$$

We now find:

$$P(C_0|x=3) = \frac{\frac{4e^{-2}}{3} \times \frac{2}{3}}{\frac{4e^{-2}}{3} \times \frac{2}{3} + \frac{120}{2^{10}} \times \frac{1}{3}} \approx 0.755$$

$$P(C_1|x=3) = \frac{\frac{120}{2^{10}} \times \frac{1}{3}}{\frac{4e^{-2}}{3} \times \frac{2}{3} + \frac{120}{2^{10}} \times \frac{1}{3}} \approx 0.245$$

Since $P(C_0|x=3) > P(C_1|x=3)$ then the test point $x=3$ is assigned to class C_0 .

```
In [1]: from scipy.special import binom
import numpy as np
```

```
y0 = 4*np.exp(-2)/3
y1 = binom(10,3)/(2**10)

p0 = 2/3
p1 = 1-p0

pos0 = (y0*p0) / (y0*p0 + y1*p1)
pos1 = (y1*p1) / (y0*p0 + y1*p1)

pos0, pos1
```

```
Out[1]: (0.7548797050821854, 0.24512029491781465)
```

Exercise 2 (7 points)

Consider the Naive Bayes classifier for a binary classification task in a two-dimensional feature space with $P(\mathbf{x}|C_1) \sim G(\mu_1, \Sigma_1)$, $P(\mathbf{x}|C_2) \sim G(\mu_2, \Sigma_2)$ and

$P(C_1) = P(C_2) = \frac{1}{2}$, where $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ is a sample observation, μ_i is the 2×1

mean vector for class C_i and Σ_i is the 2×2 covariance matrix for class C_i .

Let $g_i(x) = \ln(P(\mathbf{x}|C_i)P(C_i))$ be the discriminant function for class C_i . Then, let $g(\mathbf{x}) = g_1(\mathbf{x}) - g_2(\mathbf{x})$. The Naive Bayes decision rule is defined as:

Choose C_1 if $g(\mathbf{x}) > 0$

Choose C_2 otherwise

1. (3.5 points) Let $\mu_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $\mu_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\Sigma_1 = \Sigma_2 = \mathbf{I}$. Find $g(\mathbf{x})$. Show all your work. What is the shape of the decision boundary?

To compute the data likelihoods, we need to determine the inverse of the covariance matrix. Since they are both the identity matrix $\Sigma_1 = \Sigma_2 = \mathbf{I}$, we know that $\Sigma_1^{-1} = \Sigma_2^{-1} = \mathbf{I}^{-1} = \mathbf{I}$.

The discriminant function $g_i(x)$, $i = 1, 2$ is defined as $P(x|C_i)P(C_i)$. We can simplify it by taking $g_i(x) = \ln(P(x|C_i)P(C_i))$. Note that $x = [x_1, x_2]^T$. We have:

$$\begin{aligned} g_1(x) &= \ln \left(\frac{1}{2} \times \frac{1}{(2\pi)^{2/2} |\Sigma_1|^{1/2}} \exp \left\{ -\frac{1}{2} x^T \Sigma_1^{-1} x \right\} \right) \\ &= \ln \left(\frac{1}{2} \right) - \ln(2\pi) - \frac{1}{2} x^T x \\ &= \ln \left(\frac{1}{2} \right) - \ln(2\pi) - \frac{1}{2} (x_1^2 + x_2^2) \end{aligned}$$

and

$$\begin{aligned} g_2(x) &= \ln \left(\frac{1}{2} \times \frac{1}{(2\pi)^{2/2} |\Sigma_2|^{1/2}} \exp \left\{ -\frac{1}{2} (x - 1)^T \Sigma_2^{-1} (x - 1) \right\} \right) \\ &= \ln \left(\frac{1}{2} \right) - \ln(2\pi) - \frac{1}{2} (x - 1)^T (x - 1) \\ &= \ln \left(\frac{1}{2} \right) - \ln(2\pi) - \frac{1}{2} ((x_1 - 1)^2 + (x_2 - 1)^2) \\ &= \ln \left(\frac{1}{2} \right) - \ln(2\pi) - \frac{1}{2} (x_1^2 + x_2^2 - 2x_1 - 2x_2 + 2) \end{aligned}$$

Now, let $g(x) = g_1(x) - g_2(x)$, we find:

$$g(x) = -\frac{1}{2} (-2x_1 - 2x_2 + 2) = x_1 + x_2 - 1$$

Thus, we decide C_1 if $g(x) > 0 \iff x_2 > -x_1 + 1$. As expected, this discriminant function passes through the midpoint of the two means $[0.5, 0.5]^T$ since the two categories have the same prior and covariance matrices.

We can visualize this decision boundary as follows:

```
In [2]: import scipy.stats as stats
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
plt.style.use('bmh')

# Useful Function

def discriminant_fct(X1, X2, p1, x, g, display=True):
    # grid for plotting
    xm, ym = np.meshgrid(x, x)
```

```

X = np.flip(np.dstack([xm,ym]),axis=0)

# daata likelihoods
y1 = X1.pdf(X) #P(x|C1)
y2 = X2.pdf(X) #P(x|C2)
# visualize data likelihoods
fig =plt.figure(figsize=(12,5))
fig.add_subplot(1,2,1)
plt.imshow(y1, extent=[min(x),max(x),min(x),max(x)])
plt.colorbar(); plt.title('$P(x|C_1)$', size=20)
plt.xlabel('Feature 1'); plt.ylabel('Feature 2')
plt.xticks(fontsize=15); plt.yticks(fontsize=15)
fig.add_subplot(1,2,2)
plt.imshow(y2, extent=[min(x),max(x),min(x),max(x)])
plt.colorbar(); plt.title('$P(x|C_2)$', size=20)
plt.xlabel('Feature 1'); plt.ylabel('Feature 2')
plt.xticks(fontsize=15); plt.yticks(fontsize=15)

# priors
p2 = 1-p1
# posterior probability
pos1 = (y1*p1)/(y1*p1 + y2*p2) # P(C1|x)
pos2 = (y2*p2)/(y1*p1 + y2*p2) # P(C2|x)

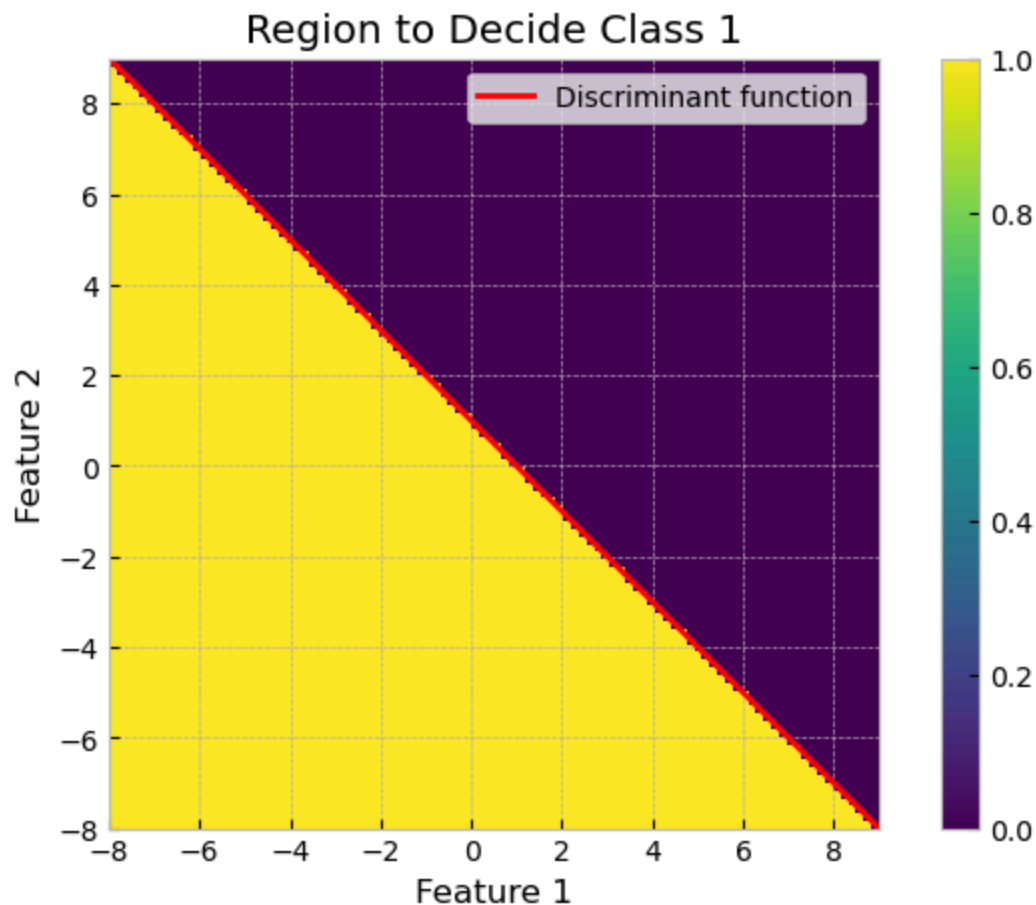
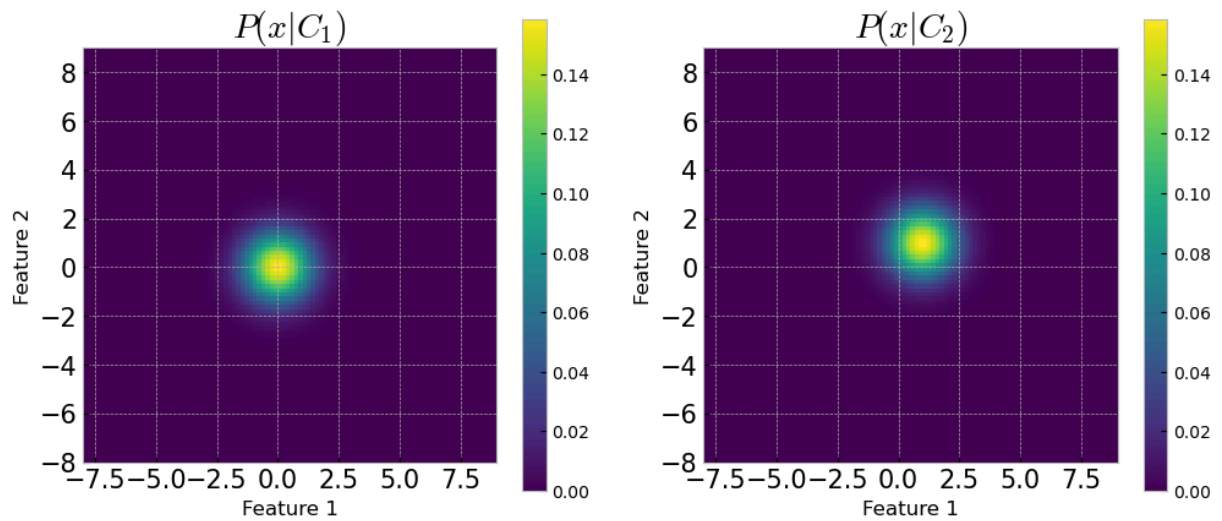
# visualize discriminant function with posterior
plt.figure(figsize=(8,5))
plt.imshow(pos1-pos2>0, extent=[min(x),max(x),min(x),max(x)])
if display:
    plt.plot(x, g, 'r', label='Discriminant function')
    plt.legend()
plt.colorbar();
plt.xlabel('Feature 1'); plt.ylabel('Feature 2')
plt.title('Region to Decide Class 1');

```

```

In [3]: X1 = stats.multivariate_normal(mean=[0,0], cov=np.eye(2))
X2 = stats.multivariate_normal(mean=[1,1], cov=np.eye(2))
p1=1/2
x = np.linspace(-8,9,100)
g = -x + 1
discriminant_fct(X1, X2, p1, x, g)

```



2. (3.5 points) Let $\mu_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $\mu_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\Sigma_1 = \begin{bmatrix} 2 & 0.5 \\ 0.5 & 2 \end{bmatrix}$ and $\Sigma_2 = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$.

Find $g(\mathbf{x})$. Show all your work. What is the shape of the decision boundary?

To compute the data likelihoods, we need to determine the inverse of the covariance matrix. We have $\Sigma_1^{-1} = \begin{bmatrix} 8/15 & -2/15 \\ -2/15 & 8/5 \end{bmatrix}$ and $\Sigma_2^{-1} = \begin{bmatrix} 5/9 & -4/9 \\ -4/9 & 5/9 \end{bmatrix}$. Moreover, the determinants are $|\Sigma_1| = \frac{15}{4}$ and $|\Sigma_2| = 9$.

The discriminant function $g_i(x)$, $i = 1, 2$ is defined as $P(x|C_i)P(C_i)$. We can simplify it by taking $g_i(x) = \ln(P(x|C_i)P(C_i))$. Note that $x = [x_1, x_2]^T$. We have:

$$\begin{aligned}
 g_1(x) &= \ln \left(\frac{1}{2} \times \frac{1}{(2\pi)^{2/2} |\Sigma_1|^{1/2}} \exp \left\{ -\frac{1}{2} x^T \Sigma_1^{-1} x \right\} \right) \\
 &= \ln \left(\frac{1}{2} \right) - \ln(2\pi) - \frac{1}{2} \ln \left(\frac{15}{4} \right) - \frac{1}{2} x^T \Sigma_1^{-1} x \\
 &= \ln \left(\frac{1}{2} \right) - \ln(2\pi) - \frac{1}{2} \ln \left(\frac{15}{4} \right) - \frac{1}{2} \left(\frac{8}{15} x_1^2 - \frac{2}{15} x_1 x_2 - \frac{2}{15} x_1 x_2 + \frac{8}{15} x_2^2 \right) \\
 &= \ln \left(\frac{1}{2} \right) - \ln(2\pi) - \frac{1}{2} \ln \left(\frac{15}{4} \right) - \frac{1}{2} \left(\frac{8}{15} x_1^2 - \frac{4}{15} x_1 x_2 + \frac{8}{15} x_2^2 \right)
 \end{aligned}$$

and

$$\begin{aligned}
 g_2(x) &= \ln \left(\frac{1}{2} \times \frac{1}{(2\pi)^{2/2} |\Sigma_2|^{1/2}} \exp \left\{ -\frac{1}{2} (x-1)^T \Sigma_2^{-1} (x-1) \right\} \right) \\
 &= \ln \left(\frac{1}{2} \right) - \ln(2\pi) - \frac{1}{2} \ln(9) - \frac{1}{2} (x-1)^T \Sigma_2^{-1} (x-1) \\
 &= \ln \left(\frac{1}{2} \right) - \ln(2\pi) - \frac{1}{2} \ln(9) - \frac{1}{2} \left(\frac{5}{9} (x_1-1)^2 - \frac{4}{9} (x_1-1)(x_2-1) - \frac{4}{9} (x_2-1)^2 \right)
 \end{aligned}$$

Let $g(x) = g_1(x) - g_2(x)$. After some simplification, we find:

$$g(x) = x_1^2 + x_2^2 - 28x_1x_2 - 10x_1 - 10x_2 + 50$$

which consists of two hyperbolas.

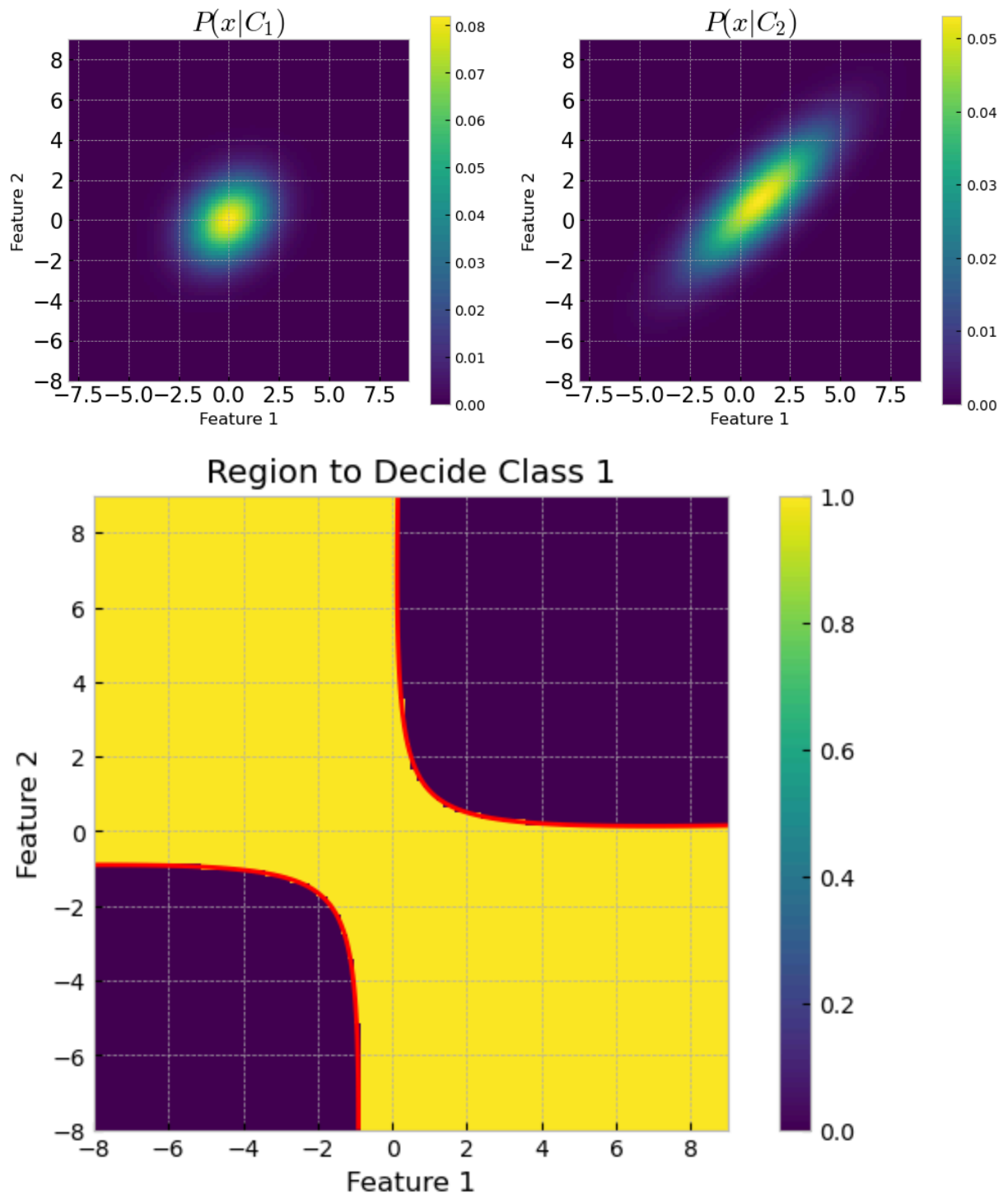
Thus, we decide C_1 if $g(x) > 0$.

We can visualize this decision boundary as follows:

```

In [4]: X1 = stats.multivariate_normal(mean=[0,0], cov=[[2,0.5],[0.5,2]])
X2 = stats.multivariate_normal(mean=[1,1], cov=[[5,4],[4,5]])
p1=1/2
x = np.linspace(-8,9,100)
x1, x2 = np.meshgrid(x, x)
discriminant_fct(X1, X2, p1, x, 0, False)
plt.contour(x1, x2, x1**2 + x2**2 -28*x1*x2 -10*x1 - 10*x2 + 50, [1], colors

```



Exercise 3 (10 points)

Use the Expectation-Maximization (EM) algorithm to solve for the parameters of an Rayleigh Mixture Model (with K components) given a set of training data $\mathbf{X} = \{x_i\}_{i=1}^N$, where $x_i \geq 0 \forall i \in \{1, \dots, N\}$. Recall the form of the Rayleigh probability density function is defined as:

$$P(x|\sigma_k) = \frac{x}{\sigma_k^2} e^{-x^2/(2\sigma_k^2)}$$

Answer the following questions:

1. (2 points) **Assuming your data is i.i.d., write down the observed data likelihood, $\mathcal{L}^0$.**

The observed data likelihood is:

$$\mathcal{L}^0 = \prod_{i=1}^N \sum_{k=1}^K \pi_k \frac{x_i}{\sigma_k^2} e^{-x_i^2/(2\sigma_k^2)}$$

2. (1 point) **Introduce the latent hidden variables $\mathbf{z}$ and write down the complete data likelihood, $\mathcal{L}^c$.**

We can introduce the hidden latent variable $\mathbf{z}$, where z_i corresponds to the Rayleigh component from which sample x_i was drawn from. Moreover, since we have a total of K Rayleigh components, $z_i \in \{1, 2, \dots, K\}$.

The complete data likelihood is given by:

$$\mathcal{L}^c = \prod_{i=1}^N \pi_{z_i} \frac{x_i}{\sigma_{z_i}^2} e^{-x_i^2/(2\sigma_{z_i}^2)}$$

3. (3 points) **Write down the EM optimization function, $Q(\Theta, \Theta^t)$, where $\Theta = \{\pi_k, \sigma_k\}_{k=1}^K$. Your final solution should contain the sum of simple (and simplified) log-terms.**

The EM optimization function is given by:

$$\begin{aligned} Q(\Theta, \Theta^t) &= \mathbb{E}_{\mathbf{z}}[\ln(\mathbf{L}^c) | \mathbf{X}, \Theta^t] \\ &= \sum_{z_i=1}^K \ln(\mathbf{L}^c) P(z_i | x_i, \Theta^t) \\ &= \sum_{k=1}^K \left[\sum_{i=1}^N \left(\ln(\pi_k) + \ln(x_i) - 2 \ln(\sigma_k) - \frac{x_i^2}{2\sigma_k^2} \right) \right] P(z_i = k | x_i, \Theta^t) \\ &= \sum_{k=1}^K \left[\sum_{i=1}^N \left(\ln(\pi_k) + \ln(x_i) - 2 \ln(\sigma_k) - \frac{x_i^2}{2\sigma_k^2} \right) \right] C_{ik} \end{aligned}$$

where $C_{ik} = P(z_i = k | x_i, \Theta^t)$.

4. (2 points) **Derive the update equations for the parameters σ_k .**

The solution for the parameter σ_k is:

$$\begin{aligned}\frac{\partial Q(\Theta, \Theta^t)}{\partial \sigma_k} &= 0 \\ \sum_{i=1}^N \left(-\frac{2}{\sigma_k} + \frac{4\sigma_k x_i^2}{(2\sigma_k^2)^2} \right) C_{ik} &= 0 \\ \sum_{i=1}^N (-2\sigma_k^2 + x_i^2) C_{ik} &= 0 \\ \sum_{i=1}^N 2\sigma_k^2 C_{ik} &= \sum_{i=1}^N x_i^2 C_{ik} \\ \sigma_k &= \sqrt{\frac{\sum_{i=1}^N x_i^2 C_{ik}}{\sum_{i=1}^N 2C_{ik}}}\end{aligned}$$

5. (2 points) **Derive the update equations for the parameters π_k .**

In order to find the solution for π_k , we must add the constraint $\sum_{k=1}^K \pi_k = 1$ to the optimization function:

$$Q_{\pi}(\Theta, \Theta^t) = Q(\Theta, \Theta^t) - \lambda \left(\sum_{k=1}^K \pi_k - 1 \right)$$

The solution for π_k is then:

$$\begin{aligned}\frac{\partial Q_{\pi}(\Theta, \Theta^t)}{\partial \pi_k} &= 0 \\ \sum_{i=1}^N \frac{1}{\pi_k} C_{ik} - \lambda &= 0 \\ \pi_k &= \frac{1}{\lambda} \sum_{i=1}^N C_{ik}\end{aligned}$$

Since $\sum_{k=1}^K \pi_k = 1$, we find that: $\sum_{k=1}^K \frac{1}{\lambda} \sum_{i=1}^N C_{ik} = 1 \iff \lambda = N$. Hence, the solution for π_k is:

$$\pi_k = \frac{\sum_{i=1}^N C_{ik}}{N}$$

Exercise 4 (18 points)

For each statement below, state whether it is true (T) or false (F) and provide a justification for your answer.

1. (2 points) **Minimizing the Least Squares objective function with a weight decay term (regularizer) is equivalent to maximizing the data likelihood with a prior on the weights.**

True. We can show this analytically. Consider the least squares objective function below with a ridge regularizer:

$$J(\mathbf{w}) = \frac{1}{2} \sum_{i=1}^N (t_i - y_i)^2 + \frac{\lambda}{2} \sum_{j=0}^M w_j^2$$

The solution for the parameters $\mathbf{w}$ is found by optimizing $J(\mathbf{w})$:

$$\begin{aligned} \arg_{\mathbf{w}} \min J(\mathbf{w}) &= \arg_{\mathbf{w}} \max -J(\mathbf{w}) \\ &= \arg_{\mathbf{w}} \max \exp(-J(\mathbf{w})) \\ &= \arg_{\mathbf{w}} \max \exp\left(-\frac{1}{2} \sum_{i=1}^N (t_i - y_i)^2 - \frac{\lambda}{2} \sum_{j=0}^M w_j^2\right) \\ &= \arg_{\mathbf{w}} \max \prod_{i=1}^N \exp\left(-\frac{1}{2} (t_i - y_i)^2\right) \prod_{j=0}^M \exp\left(-\frac{\lambda}{2} w_j^2\right) \\ &\propto \arg_{\mathbf{w}} \max \prod_{i=1}^N \mathcal{N}(t_i; y_i, 1) \prod_{j=0}^M \mathcal{N}\left(w_j; 0, \frac{1}{\lambda}\right) \\ &= \arg_{\mathbf{w}} \max p(\mathbf{t}|\mathbf{y}(\mathbf{x}; \mathbf{w}))p(\mathbf{w}|\lambda) \\ &\propto \arg_{\mathbf{w}} \max p(\mathbf{w}|\mathbf{t}) \end{aligned}$$

We see that adding a regularization penalty term to the objective function is equivalent to adding a prior probability on the parameters.

For the ridge penalty, the probabilistic model of the prior probability is a Gaussian distribution with mean 0 and variance $1/\lambda$.

2. (2 points) **The observed data likelihood on N i.i.d. samples x_i (where $x_i \geq 0, \forall i$) with a Gamma distribution, $P(x|\alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$, and a prior probability on parameter β as another Gamma distribution, $P(\beta|a, b) = \frac{b^a}{\Gamma(a)} \beta^{a-1} e^{-b\beta}$, form a Conjugate Prior relationship.**

True. The observed data likelihood for MAP is:

$$\begin{aligned}
\mathcal{L}^0 &= \left(\prod_{i=1}^N \frac{\beta^\alpha}{\Gamma(\alpha)} x_i^{\alpha-1} e^{-\beta x_i} \right) \frac{b^a}{\Gamma(a)} \beta^{a-1} e^{-b\beta} \\
&\propto \left(\prod_{i=1}^N \beta^\alpha e^{-\beta x_i} \right) \beta^{a-1} e^{-b\beta} \\
&= \beta^{\sum_{i=1}^N \alpha} e^{-\beta \sum_{i=1}^N x_i} \beta^{a-1} e^{-b\beta} \\
&= \beta^{N\alpha+a-1} e^{-\beta(\sum_{i=1}^N x_i + b)}
\end{aligned}$$

We find that the resulting posterior has the same parametric form (by a scaling factor) as the prior, thus Gamma-Gamma form a conjugate prior. The parameter updates are as follows:

$$\begin{aligned}
a^{(t+1)} &\leftarrow a^{(t)} + N\alpha \\
b^{(t+1)} &\leftarrow b^{(t)} + \sum_{i=1}^N x_i
\end{aligned}$$

3. (2 points) **In MLE estimation we are optimizing the posterior distribution.**

False. The Maximum Likelihood Estimation (MLE) finds the parameters of the observed data likelihood in a data-driven approach without any prior assumption on the parameters.

4. (2 points) **Consider $\theta_{\text{MLE}} = \arg_{\theta} \max P(X|\theta)$ and $\theta_{\text{MAP}} = \arg_{\theta} \max P(X|\theta)P(\theta)$. We have that $\theta_{\text{MLE}} = \theta_{\text{MAP}}$ for uniform-distributed priors.**

True. We can show this analytically:

$$\begin{aligned}
\theta_{\text{MAP}} &= \arg_{\mathbf{w}} \min P(X|\theta)P(\theta) \\
&= \arg_{\mathbf{w}} \min \begin{cases} P(X|\theta) \frac{1}{b-a}, & \text{if } X \text{ is continuous RV and } X \sim U(a, b) \\ P(X|\theta) \frac{1}{b-a+1}, & \text{if } X \text{ is discrete RV and } X \sim U(a, b) \end{cases}
\end{aligned}$$

since a and b are constants (in either discrete or continuous cases), then

$$\begin{aligned}
&= \arg_{\mathbf{w}} \min P(X|\theta) \\
&= \theta_{\text{MLE}}
\end{aligned}$$

5. (2 points) **The EM algorithm guarantees convergence to the global optima solution.**

False. The EM algorithm is an alternating optimization algorithm that requires a random initialization of the parameters. Different initializations will result in different solutions.

6. (2 points) **The latent variables z in the EM algorithm are always discrete random variables.**

False. The latent variables z can be discrete or continuous random variables. In the censored data example in lecture 10, the latent variables were continuous.

7. (2 points) **We can use the EM algorithm to optimize a Gamma mixture model,**

$$P(x) = \sum_{k=1}^K \pi_k \Gamma(x|\nu_k, \theta_k) \text{ to a set of } N \text{ i.i.d. samples } \{x_i\}_{i=1}^N.$$

True. Mixture models can be the sum of K components, and each component can take any shape.

8. (2 points) **The MLE solution for the parameter θ_k in the Gamma mixture model is**

$$\theta_k = \frac{\nu_k \sum_{i=1}^N C_{ik}}{\sum_{i=1}^N x_i C_{ik}}.$$

True. The observed data likelihood is:

$$\mathcal{L}^0 = \prod_{i=1}^N \sum_{k=1}^K \pi_k \Gamma(x_i|\nu_k, \theta_k) = \prod_{i=1}^N \sum_{k=1}^K \pi_k \frac{\theta_k^{\nu_k}}{\Gamma(\nu_k)} x_i^{\nu_k-1} e^{-\theta_k x_i}$$

Introducing the hidden latent variable Z , where z_i corresponds to the Gamma component from which sample x_i was drawn from. Moreover, since we have a total of K Gamma components, $z_i \in \{1, 2, \dots, K\}$.

The complete data likelihood is given by:

$$\mathcal{L}^c = \prod_{i=1}^N \pi_{z_i} \frac{\theta_{z_i}^{\nu_{z_i}}}{\Gamma(\nu_{z_i})} x_i^{\nu_{z_i}-1} e^{-\theta_{z_i} x_i}$$

The EM optimization function is given by:

$$\begin{aligned} Q(\Theta, \Theta^t) &= \mathbb{E}_z[\ln(\mathbf{L}^c) | \mathbf{X}, \Theta^t] \\ &= \sum_{z_i=1}^K \ln(\mathbf{L}^c) P(z_i | x_i, \Theta^t) \\ &= \sum_{k=1}^K \left[\sum_{i=1}^N \ln \pi_k + \nu_k \ln \theta_k - \ln \Gamma(\nu_k) + (\nu_k - 1) \ln x_i - \theta_k x_i \right] P(z_i = k | x_i, \Theta^t) \\ &= \sum_{k=1}^K \left[\sum_{i=1}^N \ln \pi_k + \nu_k \ln \theta_k - \ln \Gamma(\nu_k) + (\nu_k - 1) \ln x_i - \theta_k x_i \right] C_{ik} \end{aligned}$$

where $C_{ik} = P(z_i = k|x_i, \Theta^t)$.

The solution for the parameter θ_k is:

$$\begin{aligned}\frac{\partial Q(\Theta, \Theta^t)}{\partial \theta_k} &= 0 \\ \sum_{i=1}^N \left(\nu_k \frac{1}{\theta_k} - x_i \right) C_{ik} &= 0 \\ \frac{\nu_k}{\theta_k} \sum_{i=1}^N C_{ik} &= \sum_{i=1}^N x_i C_{ik} \\ \theta_k &= \frac{\nu_k \sum_{i=1}^N C_{ik}}{\sum_{i=1}^N x_i C_{ik}}\end{aligned}$$

9. (2 points) **A Gaussian Mixture Model can be used to find cluster groups in the data.**

True. Optimizing GMMs with the EM algorithm is an unsupervised learning approach. For a given dataset, we can learn K components that best represent the dataset. This is a form of clustering. In this context, the latent variables z correspond to the cluster labels ($z \in \{1, 2, \dots, K\}$).
